

BUS23-4

Macroeconomics

Measurement, growth, money, and stabilisation policy

Reading monetary and fiscal decisions as business signals.

Albert School — 22 hours

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1 Measuring economic activity

1.1 GDP and its three approaches

Gross domestic product (GDP) is the total market value of final goods and services produced within a country in a given period. It can be measured by three approaches, which by construction give the same total.

Definition 1 (Three approaches to GDP)

- **Production (value-added):** sum over all producers of output minus intermediate consumption, plus taxes less subsidies on products:

$$\text{GDP} = \sum_i (\text{output}_i - \text{intermediate consumption}_i) + T - S.$$

- **Expenditure:**

$$\text{GDP} = C + I + G + (X - M),$$

where C is household consumption, I investment (gross fixed capital formation plus change in inventories), G government spending, and $X - M$ net exports.

- **Income:**

$$\text{GDP} = W + \Pi + T - S,$$

the sum of compensation of employees W , gross operating surplus and mixed income Π , and taxes less subsidies on production and products.

On Eurostat data the three computations are reconciled in the national accounts; residual statistical discrepancies are recorded as an adjustment item.

1.2 Real versus nominal GDP

Nominal GDP values output at current prices; real GDP values output at the prices of a fixed base year, removing the effect of price changes.

$$\text{real GDP} = \frac{\text{nominal GDP}}{\text{deflator}} \times 100, \quad \text{deflator} = \frac{\text{nominal GDP}}{\text{real GDP}} \times 100.$$

Only real GDP measures changes in the volume of output.

Common pitfall GDP omits non-market production, the distribution of income, depletion of natural capital, and leisure. At least two welfare dimensions it fails to capture: inequality (a rise in GDP can accompany rising poverty) and environmental degradation (defensive expenditure raises GDP).

1.3 Labour-market rates

Let E be the number of employed, U the number of unemployed (without work, available, actively searching), and P the working-age population.

$$\text{labour force } L = E + U,$$

$$\text{unemployment rate} = \frac{U}{L}, \quad \text{activity (participation) rate} = \frac{L}{P}, \quad \text{employment rate} = \frac{E}{P}.$$

2 Inflation and the Phillips curve

2.1 The consumer price index

The consumer price index (CPI) tracks the cost of a fixed basket of goods and services. With base-year quantities q_i^0 (Laspeyres index):

$$\text{CPI}_t = \frac{\sum_i p_i^t q_i^0}{\sum_i p_i^0 q_i^0} \times 100, \quad \pi_t = \frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}}.$$

Causes of inflation: demand-pull (aggregate demand exceeds potential output), cost-push (rising input costs), and monetary (excessive money growth). **Costs of inflation:** menu costs, shoe-leather costs, distortion of relative prices, arbitrary redistribution between debtors and creditors, and — when unanticipated — erosion of the information content of prices. Deflation imposes the symmetric cost of rising real debt burdens.

2.2 The Phillips curve

The original Phillips curve posits a stable negative relation between unemployment u and wage (or price) inflation:

$$\pi = f(u), \quad f' < 0.$$

The Friedman–Phelps **expectations-augmented** form adds expected inflation π^e and a natural rate u^n :

$$\pi = \pi^e - \beta(u - u^n), \quad \beta > 0.$$

In the long run $\pi = \pi^e$, so $u = u^n$: there is no permanent unemployment–inflation trade-off (the long-run Phillips curve is vertical). The **NAIRU** (non-accelerating-inflation rate of unemployment) is the rate u^n at which inflation is stable.

Common pitfall Empirical limits: the stable original curve broke down in the 1970s stagflation (high inflation with high unemployment); the NAIRU is unobservable and estimated with wide uncertainty; and the curve has appeared flat in recent decades, weakening its use for policy.

3 Economic growth

3.1 Growth accounting and total factor productivity

With an aggregate production function $Y = AF(K, L)$, growth accounting decomposes output growth into contributions from capital, labour, and total factor productivity (TFP), the Solow residual:

$$\frac{\dot{Y}}{Y} = \frac{\dot{A}}{A} + \alpha \frac{\dot{K}}{K} + (1 - \alpha) \frac{\dot{L}}{L},$$

where α is the elasticity of output with respect to capital (its income share under competitive markets). TFP growth is computed as the residual:

$$\frac{\dot{A}}{A} = \frac{\dot{Y}}{Y} - \alpha \frac{\dot{K}}{K} - (1 - \alpha) \frac{\dot{L}}{L}.$$

3.2 The Solow model

Take Cobb–Douglas $Y = AK^\alpha L^{1-\alpha}$ with $0 < \alpha < 1$, saving rate s , depreciation δ , labour-force growth n . In per-worker terms $y = Ak^\alpha$ with $k = K/L$, capital accumulates as

$$\dot{k} = sy - (n + \delta)k = sAk^\alpha - (n + \delta)k.$$

Proposition 2 (Steady state) The steady state $\dot{k} = 0$ gives

$$k^* = \left(\frac{sA}{n + \delta} \right)^{\frac{1}{1-\alpha}}, \quad y^* = A(k^*)^\alpha.$$

Per-worker output is constant in steady state; with TFP growth g , output per worker grows at g .

Proposition 3 (Golden rule) The saving rate maximising steady-state consumption $c^* = (1 - s)y^*$ satisfies the golden-rule condition that the marginal product of capital equals $n + \delta$:

$$\alpha A(k_{\text{gold}})^{\alpha-1} = n + \delta.$$

Conditional convergence: because the marginal product of capital diminishes, economies with the same parameters (s, n, δ, A) converge to the same steady state, and those farther below it grow faster. Convergence is conditional on sharing these determinants, not absolute.

4 Endogenous growth and sustainability

In the Solow model long-run growth comes from exogenous TFP. **Endogenous growth** theory makes the growth rate result from choices within the model.

- **Romer-style innovation:** ideas are non-rival; R&D produces new designs, and increasing returns to knowledge sustain growth without a vanishing return.
- **Human capital:** treating education and skills as accumulable capital H enters the production function and raises long-run output.
- **Secular stagnation:** the hypothesis that chronically deficient demand or a low natural rate of interest holds growth persistently below potential.

4.1 Sustainability constraints

The **Kaya identity** decomposes carbon emissions:

$$\text{CO}_2 = \text{Pop} \times \frac{\text{GDP}}{\text{Pop}} \times \frac{\text{Energy}}{\text{GDP}} \times \frac{\text{CO}_2}{\text{Energy}},$$

i.e. population, income per capita, energy intensity, and carbon intensity. **Relative decoupling** is a fall in emissions per unit of GDP; **absolute decoupling** is a fall in total emissions while GDP rises. Only absolute decoupling reduces emissions in level.

5 Money and the quantity theory

5.1 Endogenous money creation

In a modern banking system money is created when commercial banks extend credit: a new loan simultaneously creates a deposit. Lending is constrained by capital requirements, the demand for credit, and the policy rate rather than by a fixed stock of reserves. The **credit (deposit) multiplier** with reserve ratio r gives an upper bound on deposits from base money B :

$$D = \frac{B}{r}.$$

5.2 Monetary aggregates

- **M1:** currency in circulation plus overnight deposits.
- **M2:** M1 plus deposits with maturity up to two years and redeemable at notice up to three months.
- **M3:** M2 plus repurchase agreements, money-market fund shares, and debt securities up to two years.

5.3 The quantity theory of money

Definition 4 (Quantity equation)

$$MV = PY,$$

with M money stock, V velocity, P the price level, Y real output. In growth rates,

$$\frac{\dot{M}}{M} + \frac{\dot{V}}{V} = \frac{\dot{P}}{P} + \frac{\dot{Y}}{Y}.$$

Assuming V constant and Y at potential, money growth maps one-for-one into inflation. **Limits:** velocity is not stable in the short run (notably after 2008, when large increases in M did not

produce proportional inflation), and the direction of causation is contested; the theory holds better as a long-run, high-inflation approximation.

6 Central banking

A central bank's **mandate** is typically price stability (the ECB targets 2% inflation over the medium term), sometimes alongside maximum employment (the Fed's dual mandate). **Independence** — operational separation from the government — is intended to make a low-inflation commitment credible.

Conventional instruments: the policy interest rate, open-market operations, the reserve requirement, and standing facilities.

Unconventional tools, used when the policy rate is at its effective lower bound:

- **Quantitative easing (QE):** large-scale purchases of assets to lower long-term yields and expand the balance sheet.
- **Forward guidance:** communication about the future path of policy to steer expectations.

Comparative 2022–2024: facing the post-pandemic inflation shock, the Fed and the ECB raised policy rates sharply from 2022 and began reducing their balance sheets; the SNB raised its rate, exiting negative territory in 2022, while continuing to use foreign-exchange interventions given the franc's safe-haven role.

7 Fiscal policy

7.1 The Keynesian multiplier

With consumption $C = c_0 + c(Y - T)$ where c is the marginal propensity to consume ($0 < c < 1$), equilibrium output $Y = C + I + G$ with lump-sum tax T solves to

$$Y = \frac{1}{1-c}(c_0 - cT + I + G).$$

Proposition 5 (Spending and tax multipliers)

$$\frac{\partial Y}{\partial G} = \frac{1}{1-c}, \quad \frac{\partial Y}{\partial T} = \frac{-c}{1-c}.$$

Numerical example: with $c = 0.8$, the spending multiplier is $\frac{1}{1-0.8} = 5$, so $\Delta G = 10$ raises output by 50; the tax multiplier is $-\frac{0.8}{0.2} = -4$.

Automatic stabilisers (progressive taxes, unemployment benefits) dampen the cycle without new legislation, whereas **discretionary policy** requires deliberate decisions and is subject to recognition, decision, and implementation lags.

Crowding-out: a debt-financed rise in G raises the interest rate, which reduces private investment, partly offsetting the stimulus — the effect made explicit in IS-LM below.

8 IS–LM

The IS–LM model determines output Y and the interest rate i jointly in goods and money markets, with the price level fixed.

Definition 6 (The two curves)

- **IS** (goods market): equilibrium $Y = C(Y - T) + I(i) + G$ with $I' < 0$ gives a **downward-sloping** relation between Y and i .
- **LM** (money market): equilibrium $M/P = L(Y, i)$ with $L_Y > 0$, $L_i < 0$ gives an **upward-sloping** relation between Y and i .

Equilibrium is their intersection.

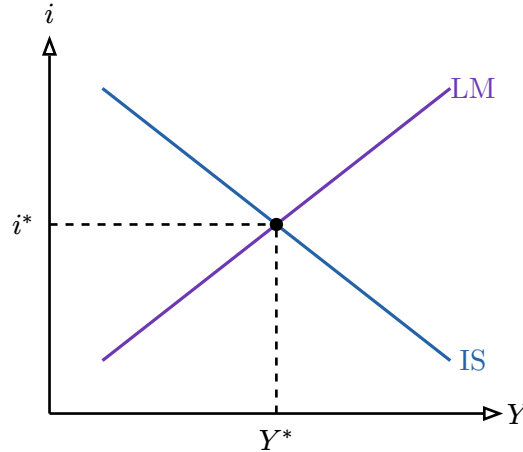


Figure 1: IS–LM equilibrium at the intersection of the goods-market (IS) and money-market (LM) curves.

Shifts. Expansionary fiscal policy (higher G or lower T) shifts IS right, raising both Y and i . Expansionary monetary policy (higher M) shifts LM right, raising Y and lowering i .

Gaps. Let Y_n be potential output. If equilibrium $Y^* < Y_n$ there is a **recessionary gap**; if $Y^* > Y_n$ there is an **inflationary gap**. The position of the curves relative to Y_n identifies the gap, and the required policy is the shift that moves Y^* to Y_n .

Worked example — Algebraic solution. IS: $Y = c_0 + c(Y - T) + (b_0 - bi) + G$; LM: $M/P = d_0 + d_1Y - d_2i$. Solving the two linear equations for i and substituting back yields closed forms for i^* and Y^* in terms of the parameters; crowding-out appears as the $-bi$ term reducing the pure multiplier $1/(1 - c)$.

9 Debt/GDP dynamics

Let $b = B/Y$ be the public-debt-to-GDP ratio, d the primary deficit ratio, r the real interest rate on debt, g the real growth rate. The fundamental debt dynamics equation is

$$\Delta b \approx (r - g)b + d.$$

Proposition 7 (Stabilising the ratio) When $r > g$, a primary surplus ($d < 0$) is required merely to stabilise b ; the snowball term $(r - g)b$ then raises debt on its own. When $g > r$, the ratio falls even with a small primary deficit.

Critical thresholds (e.g. the 60%-of-GDP Maastricht reference) are conventions rather than precise tipping points; sustainability depends on the sign of $r - g$ and the credibility of future primary balances. On real data, 2008 and 2020 produced large jumps in b through falling Y and rising deficits.

10 Policy-mix case studies

Episode	Shock	Monetary response	Fiscal response
2008 GFC	Financial collapse, credit freeze, demand slump	Rates cut to the lower bound; QE launched; lender of last resort	Stimulus packages, bank rescues; later austerity in the euro area
2020 COVID	Supply-and-demand shock from lockdowns	Rates low; large QE; liquidity facilities	Furlough/short-time schemes, transfers, guarantees — very large deficits
2022 inflation	Energy and post-pandemic inflation surge	Rapid rate hikes; balance-sheet reduction	Energy-price support; tension with monetary tightening

Evaluation. In 2008 monetary easing prevented collapse, but euro-area austerity (with $r > g$) is widely judged to have prolonged the recession. In 2020 coordinated monetary–fiscal expansion supported incomes, at the cost of debt ratios and contributing, with supply shocks, to the 2022 inflation. In 2022 tightening reduced inflation but raised financing costs — the direct business signal: higher policy rates raise the WACC used to discount investment and raise the financing cost of working capital (BFR), the link carried into BUS23-3 and BUS33-3.